

Anxin (Bob) Guo

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Education

Northwestern University, Evanston, Illinois

September 2023 – Present

Ph.D. student in Computer Science.

Northwestern University, Evanston, Illinois

September 2019 – June 2023

B.A. in Mathematics and M.S. in Computer Science. GPA: 3.92/4.00.

Research Interests

Theoretical computer science and the theoretical foundations of machine learning, including learning theory, neural network expressivity, and information-theoretic interpretations of language models.

Research Projects and Papers

Manuscripts and Ongoing Projects

On Hint-Induced Answer Shifts in Chain-of-Thought Reasoning

Sara Ahmadian, Pranjal Awasthi, **Anxin Guo**, Kamesh Munagala, Chirag Pabbaraju, and Aravindan Vijayaraghavan. (alphabetical order)

Manuscript in preparation.

Algorithmic Lemma Discovery via Formal Library Compression

Ruize Xu, Yi Gu, **Anxin Guo**, Tianchong Jiang, Chenxiao Yang, and Zhiyuan Li.

Manuscript.

Copying Under Repetition: From Prefix Matching to Position Hashing

Anxin Guo, Tianlong Huang, Chenxiao Yang, and Zhiyuan Li.

Manuscript.

- Formalized finite-horizon uniformity, an expressivity regime between non-uniform circuits and fully uniform transformers, where one model must handle all inputs up to length N while resources may scale with N .
- Proved impossibility results for a class of transformers spanning NoPE and a practical implementation of ALiBi: they cannot correctly terminate the copy task even with depth $N^{0.01}$, near-logarithmic precision, and arbitrary width.
- Designed a modular attention head, a plug-in augmentation using value rotation with integer period p to encode query-key distance through the value stream.
- Showed that the augmented architecture can copy arbitrary strings using logarithmic width and $\log \log N$ precision.
- Built diagnostic experiments showing that frontier 400B-parameter language models struggle to copy strings with extreme repetition.

Publications

Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem for Membership Testing

Anxin Guo and [Jingwei Li](#).

International Conference on Machine Learning (ICML), 2026, spotlight, top 2.2%. [arXiv](#).

- Defined membership testing as a unifying abstraction for approximate set membership, Bloom-filter-style error metrics, and factual errors in language models.
- Established the optimal space-error tradeoff in this framework, with an asymptotic lower bound governed

by relative entropy, analogous to rate-distortion theory.

- Showed that optimally trained language models under cross-entropy loss with bounded memory may assign high confidence to unseen non-facts as a structural consequence of limited capacity.
- Extended known bounds for Bloom-type filters into a tight form, including regimes where false negatives are allowed.

Agnostic Learning of Arbitrary ReLU Activation under Gaussian Marginals

Anxin Guo and [Aravindan Vijayaraghavan](#).

Conference on Learning Theory (COLT), 2025. [arXiv](#); [conference version](#); [recorded talk](#).

- Gave the first polynomial-time constant-factor approximation agnostic learner for arbitrarily biased ReLU neurons under Gaussian marginals.
- Established a formal separation between Statistical Query (SQ) and Correlational Statistical Query (CSQ) models for this problem, proving limitations of gradient-based algorithms in this setting.
- Designed a method combining Thresholded PCA and Reweighted Gradient Descent to overcome limitations of standard gradient-based approaches.

To Store or Not to Store: A Graph-Theoretic Approach for Dataset Versioning

Anxin Guo, [Jingwei Li](#), [Pattara Sukprasert](#), [Samir Khuller](#), [Amol Deshpande](#), and [Koyel Mukherjee](#).

IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024. [arXiv](#); [conference version](#).

- Formulated and analyzed approximation algorithms and heuristics for optimizing the storage-retrieval cost tradeoff in large-scale dataset versioning systems.
- Designed provably near-optimal algorithms for tree-like graphs by exploiting low-treewidth structure.
- Implemented and benchmarked proposed heuristics on experiments based on real-world GitHub repository data, improving performance by up to 1000x over prior methods.

Awards and Honors

Ph.D. Student Research Award, Computer Science Department, Northwestern University 2024 – 2025

Barris Award for Outstanding TA, CS 335: Introduction to the Theory of Computing Fall 2024

Junior Career Award in Mathematics, Mathematics Department, Northwestern University 2021 – 2022

Research and Teaching

IDEAL-funded summer research exchange, University of Chicago Summer 2026

Hosted by [Chao Gao](#).

Research Intern / IDEAL-funded summer research exchange, TTIC Summer 2025

Toyota Technological Institute at Chicago. Hosted by [Zhiyuan Li](#).

- Initiated the transformer copying project during the internship.
- Built experiments validating the failure mode of frontier language models and the proposed architecture.
- Presented at a reading group.

Teaching Assistant, CS 336: Design and Analysis of Algorithms Spring 2026

Teaching Assistant, CS 335: Introduction to the Theory of Computing Fall 2024

- Gave a guest lecture on busy beaver numbers in addition to regular TA duties.
- Awarded the Barris Award for Outstanding TA.

Service and Programs

Program Committee / Reviewer, Reliable ML from Unreliable Data, NeurIPS Workshop 2025

Volunteer, 65th IEEE Symposium on Foundations of Computer Science (FOCS) 2024

Ross Mathematics Program (Asia), student, junior counselor, counselor
Directed Reading Program, Linear Representations of Finite Groups
Read with [Wenyuan Li](#).

2018 – 2020
Spring 2021

Skills

Technology: C++, Python (NumPy, PyTorch, pandas, matplotlib), Mathematica, and Gurobi Optimizer.

Writing

I have written three Chinese-language articles on theoretical computer science for [my Zhihu column](#).